

Requirements Document for Customizing Pico VR Headset Software for Omni One System

Project Overview

Virtuix aims to customize the Pico VR headset software to seamlessly integrate with the Omni One treadmill system. The primary feature to be adjusted is the “play boundary” setup app, which needs to accommodate both the unique constraints of the Omni One treadmill and the flexibility for users to switch to room-scale mode when using the headset independently of the treadmill.

Purpose

This document outlines the requirements for customizing the Pico VR headset software to integrate with Virtuix's Omni One treadmill system, enhancing user experience by streamlining the play boundary setup and ensuring seamless mode switching.

Scope

This project focuses on modifying the Pico boundary system to support both Omni One mode and traditional room-scale mode. The scope includes:

- Automatic detection when using the Omni One treadmill.
- Provision for users to switch to and from room-scale mode.
- Development of user interfaces to facilitate seamless mode switching and user profile management.

Key Objectives

Implement Omni One Mode

- **Objective:** Eliminate the need for users to draw a boundary when using the Omni One treadmill.

- **Target:** Achieve automatic boundary detection with a success rate of 99% upon system startup.

Maintain Room-Scale Mode

- **Objective:** Allow users to switch to traditional room-scale VR for games that do not require the treadmill.
- **Target:** Ensure seamless activation of room-scale mode with boundary setup completed within 5 minutes for 95% of users.

Seamless Mode Switching

- **Objective:** Ensure an intuitive and straightforward process for switching between Omni One mode and room-scale mode.
- **Target:** Enable mode switching within 30 seconds with a success rate of 98%, minimizing user errors and confusion.

Background

Context

Virtuix, known for its Omni One VR treadmill, aims to provide an immersive VR experience that allows full-body movement. Traditional VR headsets like the Pico require users to manually draw a play boundary, which is redundant for Omni One users confined within the treadmill.

Existing System

The current Pico VR headset boundary setup involves user-drawn boundaries for safe VR play. This process is unnecessary for Omni One users, requiring a customized solution to eliminate redundancy and enhance user experience.

Assumptions and Dependencies

Assumptions

- Users have basic familiarity with VR boundary setup procedures.
- Omni One treadmills are correctly installed and calibrated before use.
- Users will utilize both the treadmill-specific and room-scale functionalities as needed.
- Users will follow safety guidelines provided by Virtuix for both modes of operation.

Dependencies

- Integration with Omni Pro SDK for real-time communication between the treadmill and the headset is crucial. Any delays or changes in the SDK could impact the project timeline.
- Collaboration with Pico to ensure software modifications align with headset capabilities. This includes timely updates and support from Pico.
- Reliable network connectivity to ensure real-time data exchange between the Omni One treadmill and the VR headset. Network issues could affect system performance and user experience.
- Compliance with relevant VR industry standards and safety regulations to ensure user safety and data protection.

Requirements

1. Omni One Mode

- **Automatic Boundary Detection:**
 - **Functionality:** Upon startup in Omni One mode, the system should automatically detect that it is connected to an Omni One treadmill.
 - **Boundary Configuration:** Disable the manual boundary drawing process. Instead, load a predefined, non-changeable boundary that corresponds to the Omni One treadmill area.
 - **User Confirmation:** Include a simple confirmation screen indicating that Omni One mode is active and explaining the boundary setup is unnecessary.
- **Safety Features:**
 - **Alert System:** Implement an alert system that notifies users if they are attempting to move outside the treadmill area, reducing the risk of accidents.
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2. Seamless Mode Switching

- **Mode Toggle:**
 - **Interface:** Create a user-friendly interface within the VR headset settings menu to toggle between Omni One mode and room-scale mode.
 - **Confirmation Prompt:** Include a confirmation prompt to prevent accidental switching between modes.

- **Instructions:** Provide clear, concise instructions on how to switch modes, including the implications for boundary setup and user movement.
- **Profile Management:**
 - **User Profiles:** Implement user profiles that save preferences for both Omni One and room-scale modes, ensuring a personalized experience.
 - **Settings Synchronization:** Synchronize settings across multiple devices if users use their Pico VR headset with different systems.

3. Integration with Omni One Treadmill

- **API Communication:**
 - **Real-Time Data Exchange:** Ensure the VR headset communicates in real-time with the Omni One treadmill for accurate boundary and user movement tracking.
 - **SDK Utilization:** Utilize the Omni Pro SDK to integrate treadmill-specific functionalities such as speed, direction, and positional tracking.
- **Calibration Process:**
 - **Initial Setup:** Include a one-time calibration process during the initial setup to align the headset's tracking system with the Omni One treadmill.

4. Testing and Validation

- **Beta Testing:**
 - **Internal Testing:** Conduct extensive internal testing with a variety of scenarios to ensure the new boundary setup system works flawlessly in both modes.
 - **User Feedback:** Gather feedback from a closed beta group of Omni One users to identify and address any usability issues or bugs.
- **Compatibility Check:**
 - **Game Compatibility:** Test the new system with a wide range of VR games to ensure compatibility and optimal performance.

Non-functional Requirements

Performance

Ensure real-time data exchange between the headset and the Omni One treadmill with a latency of no more than 50 milliseconds to provide a seamless user experience. The system should handle peak loads without performance degradation.

Usability

Maintain a high standard for user safety and interface intuitiveness. The setup process should be completed by 95% of users within 2 minutes for Omni One mode and within 5 minutes for room-scale mode.

Reliability

The system must consistently detect and switch modes with a failure rate of less than 1%. Automatic boundary detection and mode switching should operate correctly under various environmental conditions and usage scenarios.

Security

Ensure user data and profile settings are securely managed and stored, complying with GDPR and other relevant data protection regulations. Use encryption for data at rest and in transit, and implement robust access controls.

User Stories or Use Cases

Use Case 1: User starts the VR headset in Omni One mode.

- **Preconditions:** Omni One treadmill is connected and calibrated.
- **Steps:** System detects treadmill, loads Omni One boundary, confirms mode activation to the user.
- **Postconditions:** User is informed that Omni One mode is active and can begin VR session without manual boundary setup.

Use Case 2: User switches to room-scale mode.

- **Preconditions:** User is in Omni One mode, headset is operational.
- **Steps:** User accesses settings menu, toggles to room-scale mode, draws boundary using standard process.
- **Postconditions:** Boundary is set, and the user can engage in room-scale VR gameplay.

Use Case 3: User saves and retrieves boundary settings.

- **Preconditions:** User has previously used both Omni One and room-scale modes.

- **Steps:** User toggles mode, system recalls last used boundary setup for the selected mode.
- **Postconditions:** User seamlessly switches between VR environments with previously saved boundary settings.

Use Case 4: User encounters a boundary violation alert.

- **Preconditions:** User is in Omni One mode and moves towards the edge of the treadmill area.
- **Steps:** System detects potential boundary breach, triggers alert to notify user.
- **Postconditions:** User is warned and adjusts position, ensuring safety within the designated area.

User Interface and Design Requirements

Design Mockups

- **Omni One Mode Confirmation Screen:** A simple interface that informs users of the Omni One mode activation, explaining that the boundary setup is predefined and highlighting the safety features.
- **Mode Toggle Interface:** A settings menu with a clear toggle switch between Omni One and room-scale modes, featuring intuitive icons and clear labels.
- **Boundary Setup Interface:** For room-scale mode, a step-by-step guide to drawing the play boundary, supported by visual aids and progress indicators.

Interface Guidelines

- **Consistency:** Ensure a consistent look and feel across all UI elements, aligning with Virtuix's branding and design standards.
- **Intuitiveness:** Design the interface to be intuitive, minimizing the learning curve for new users. Use familiar symbols and straightforward language.
- **Accessibility:** Ensure the UI is accessible, incorporating features such as voice instructions and adjustable text sizes to cater to a wide range of users.
- **Feedback Mechanisms:** Include real-time feedback to guide users through the setup process, confirming actions and providing tips or warnings as needed.

Product Metrics

Success Criteria

- **User Adoption Rates:** Time spent in each mode, sessions in each mode, average session length in each mode.
- **Setup Time Reduction:** Track the average time taken by users to set up the boundary, aiming to reduce this time by 50% in Omni One mode compared to traditional room-scale setup times.
- **User Satisfaction Scores:** Collect user feedback post-implementation, targeting an average satisfaction rating of 8 out of 10 or higher regarding the new setup process. Eg. NPS

Performance Metrics

- **System Responsiveness:** Monitor the time taken for the system to detect the Omni One treadmill and switch modes, aiming for a response time of less than 50 milliseconds.
- **Mode Switching Success Rate:** Track the frequency of successful mode switches without user-reported issues or errors, aiming for a success rate of 98%.

Constraints and Limitations

Technical Constraints

- **Hardware Limitations:** The customization must operate within the existing capabilities of the Pico VR headset hardware, without necessitating hardware upgrades.
- **Connectivity Requirements:** Reliable real-time communication between the headset and the Omni One treadmill is essential for accurate boundary detection and user movement tracking.

Regulatory Constraints

- **Compliance:** Ensure the system complies with VR industry standards and safety regulations, including guidelines for user safety in immersive environments.
- **Data Privacy:** Adhere to data privacy regulations, ensuring secure handling and storage of user data and profile settings.

Acceptance Criteria

- **Mode Switching Reliability:** Users should be able to switch between modes with a high success rate and minimal issues.
- **User Feedback:** Gather user feedback indicating a high level of satisfaction with the new boundary setup process and ease of use.
- **Safety Features:** The alert system must effectively notify users of boundary violations, ensuring they remain within the safe operational area of the Omni One treadmill.

Timeline and Milestones

1. **Project Initiation and Planning (Q3 2024)**
 - **Start Date:** July 1, 2024
 - **Activities:** Define project scope, create a detailed project plan, allocate resources, and set up communication channels with Pico.
 - **Completion Date:** July 15, 2024
2. **Requirement Gathering and Analysis (Q3 2024)**
 - **Start Date:** July 16, 2024
 - **Activities:** Conduct meetings with stakeholders to gather detailed requirements, document functional and non-functional requirements, and finalize the PRD.
 - **Completion Date:** August 15, 2024
3. **Design Phase (Q3 2024)**
 - **Start Date:** August 16, 2024
 - **Activities:** Create design mockups for the user interface, develop wireframes, and design system architecture for integrating the new boundary system.
 - **Completion Date:** September 15, 2024
4. **Development Phase (Q4 2024)**
 - **Start Date:** September 16, 2024
 - **Activities:** Mode switching interface, and profile management. Collaborate closely with Pico's development team for software modifications.
 - **Completion Date:** November 15, 2024
5. **Internal Testing and QA (Q4 2024)**
 - **Start Date:** November 16, 2024
 - **Activities:** Conduct extensive internal testing to identify and fix bugs, perform usability testing, and ensure the system meets all functional and non-functional requirements.
 - **Completion Date:** December 15, 2024
6. **Beta Release and Testing (Q1 2025)**

- **Start Date:** January 2, 2025
- **Activities:** Release beta version to a selected group of testers, gather feedback, identify issues, and make necessary adjustments based on user feedback.
- **Completion Date:** February 15, 2025
- 7. **Final Adjustments and Optimization (Q1 2025)**
 - **Start Date:** February 16, 2025
 - **Activities:** Implement final adjustments based on beta testing feedback, optimize system performance, and conduct final validation tests.
 - **Completion Date:** March 15, 2025
- 8. **Final Testing and Validation (Q2 2025)**
 - **Start Date:** March 16, 2025
 - **Activities:** Perform comprehensive testing including stress tests, integration tests, and final user acceptance testing to ensure the product is ready for launch.
 - **Completion Date:** April 15, 2025
- 9. **Full Deployment (Q2 2025)**
 - **Start Date:** April 16, 2025
 - **Activities:** Roll out the final version of the customized Pico VR headset software to all Omni One users, provide training materials, and offer customer support.
 - **Completion Date:** May 15, 2025
- 10. **Post-Launch Support and Monitoring (Q2 2025)**
 - **Start Date:** May 16, 2025
 - **Activities:** Monitor system performance, address any post-launch issues, and gather user feedback for continuous improvement.
 - **Completion Date:** June 30, 2025

Summary Timeline

- **Project Initiation and Planning:** July 1, 2024 – July 15, 2024
- **Requirement Gathering and Analysis:** July 16, 2024 – August 15, 2024
- **Design Phase:** August 16, 2024 – September 15, 2024
- **Development Phase:** September 16, 2024 – November 15, 2024
- **Internal Testing and QA:** November 16, 2024 – December 15, 2024
- **Beta Release and Testing:** January 2, 2025 – February 15, 2025
- **Final Adjustments and Optimization:** February 16, 2025 – March 15, 2025
- **Final Testing and Validation:** March 16, 2025 – April 15, 2025
- **Full Deployment:** April 16, 2025 – May 15, 2025
- **Post-Launch Support and Monitoring:** May 16, 2025 – June 30, 2025